

THE LIMESTONE, Australia

WMO: 948130

Lat: 36.9656S

Lon: 139.7164E

Elev: 18

StdP: 101.11

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	5.2	0.9	4.0	10.1	2.0	4.4	9.5	15.6	11.5	14.4	11.3	3.9	70	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	32.1	18.0	29.2	17.7	26.5	17.0	19.9	27.2	18.9	25.4	18.1	23.9	6.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.8	21.3	16.9	12.1	20.2	16.1	11.5	19.6	57.0	27.1	53.7	25.4	51.2	23.9	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	12.4	11.4	2.1	37.0	0.8	2.2	1.5	38.6	1.0	39.8	0.6	41.1	0.0	42.6		
			1.3	21.9	0.9	1.1	0.6	22.7	0.1	23.4	-0.4	24.0	-1.1	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	18.3	18.5	17.1	14.9	13.0	11.2	10.7	11.0	12.1	13.5	15.4	17.0
	DBStd	3.75	3.37	3.24	3.04	2.38	2.05	1.87	1.73	1.70	1.92	2.54	3.07	3.15
	HDD10.0	36	0	0	0	0	2	9	13	9	3	1	0	0
	HDD18.3	1592	40	33	63	106	166	214	236	227	187	152	102	66
	CDD10.0	1631	257	237	219	148	94	45	36	41	66	110	162	217
	CDD18.3	144	39	36	24	4	1	0	0	0	0	3	13	24
	CDH23.3	1417	404	312	199	34	2	0	0	0	1	44	152	269
CDH26.7	559	183	125	70	6	0	0	0	0	0	10	51	114	
Wind	WSAvg	6.1	6.6	6.3	5.8	5.3	5.5	5.7	6.2	6.2	6.3	6.2	6.3	6.3
Precipitation	PrecAvg	551	18	18	20	37	61	82	93	73	57	34	26	26
	PrecMax	777	59	59	68	68	126	125	155	131	109	68	74	102
	PrecMin	347	1	0	2	9	14	28	42	22	30	6	3	5
	PrecStd	107	15	17	14	15	27	27	35	34	22	20	16	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	34.5	32.2	27.0	21.4	16.8	15.9	18.2	21.5	27.9	31.6	34.3
		MCWB	19.0	19.4	17.5	16.1	14.5	13.9	13.2	13.0	14.6	16.1	17.4	18.2
	2%	DB	31.8	30.6	28.4	23.5	18.7	15.6	14.7	16.0	18.7	23.8	27.6	29.9
		MCWB	18.1	18.4	17.4	15.2	14.4	13.1	12.2	12.7	13.8	15.2	16.7	17.2
	5%	DB	27.8	26.9	25.0	21.0	17.4	14.9	14.1	14.8	17.0	20.7	24.3	26.0
		MCWB	17.7	17.9	16.5	15.1	14.2	12.7	11.8	12.3	13.4	14.3	16.0	16.5
	10%	DB	24.3	24.1	22.1	19.0	16.4	14.2	13.5	14.0	15.6	18.3	21.2	22.6
		MCWB	17.2	17.5	16.1	14.8	13.8	12.1	11.4	11.6	12.7	13.8	15.2	16.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.7	19.7	17.8	16.3	15.1	13.8	14.2	16.5	17.2	19.4	20.3
		MCDB	28.4	27.4	26.8	21.9	17.9	15.9	15.2	16.0	18.8	23.8	27.0	27.2
	2%	WB	19.7	19.9	18.5	16.9	15.5	13.8	12.9	13.3	14.8	16.0	17.8	18.6
		MCDB	27.5	26.4	24.4	20.2	17.2	14.9	14.1	15.1	17.5	21.2	24.3	25.6
	5%	WB	18.7	18.8	17.6	16.1	14.8	13.1	12.3	12.6	13.9	15.0	16.7	17.6
		MCDB	25.0	24.8	22.8	19.2	16.7	14.5	13.6	14.4	16.2	19.5	22.5	23.7
	10%	WB	17.7	17.8	16.8	15.3	14.1	12.5	11.7	12.0	13.1	14.1	15.6	16.7
		MCDB	23.2	22.9	21.2	18.4	16.1	14.0	13.2	13.6	15.2	17.9	20.7	22.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.1	8.5	7.9	6.2	5.6	5.1	5.5	6.5	8.3	9.2	9.1
		MCDBR	16.0	14.3	13.8	11.6	7.9	6.0	5.2	6.8	9.5	14.1	15.6	16.2
		MCWBR	5.7	5.5	5.7	5.8	5.2	4.8	4.2	4.9	6.1	7.4	6.7	6.2
	5% WB	MCDBR	11.8	11.4	10.3	8.3	6.3	5.2	4.8	6.2	7.5	11.2	12.8	12.1
		MCWBR	5.5	5.2	5.2	5.3	4.8	4.5	4.3	4.8	5.5	6.5	6.2	5.7
Clear-Sky Solar Irradiance	taub	0.340	0.337	0.323	0.317	0.307	0.304	0.308	0.317	0.337	0.355	0.338	0.336	
	taud	2.538	2.558	2.597	2.602	2.612	2.601	2.585	2.550	2.484	2.429	2.512	2.539	
	Ebn at Noon	993	974	950	896	847	819	836	881	919	944	989	1002	
	Edh at Noon	109	102	91	80	69	65	70	82	100	115	111	110	
All-Sky Solar Radiation	RadAvg	7.56	6.56	5.21	3.63	2.43	1.98	2.14	3.05	4.32	5.70	6.76	7.46	
	RadStd	0.36	0.33	0.21	0.23	0.13	0.12	0.11	0.16	0.35	0.37	0.27	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-111	+132	N/A
(47) Regional (14 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-114	+136	+36

Nomenclature: See separate page